the magnitude of the velocity is called v, then of course

$$\Delta v_\perp = v\Delta\theta$$

and the acceleration a will be

$$a_\perp = v(\Delta\theta/\Delta t).$$

Now we need to know $\Delta\theta/\Delta t$, which can be found this way: If, at the given moment, the curve is approximated as a circle of a certain radius R, then in a time Δt the distance s is, of course, $v\,\Delta t$, where v is the speed.

$$\Delta\theta = (v\Delta t)/R, \quad \text{or} \quad \Delta\theta/\Delta t = v/R.$$

Therefore, we find

$$a_\perp = v^2/R, \tag{1.16}$$

as we have seen before.

1-7 Scalar product of vectors

Now let us examine a little further the properties of vectors. It is easy to see that the *length* of a step in space would be the same in any coordinate system. That is, if a particular step $\mathbf{r}$ is represented by x, y, z, in one coordinate system, and by x', y', z' in another coordinate system, surely the distance $r = |\mathbf{r}|$ would be the same in both. Now

$$r = \sqrt{x^2 + y^2 + z^2}$$

and also

$$r' = \sqrt{x'^2 + y'^2 + z'^2}.$$

So what we wish to verify is that these two quantities are equal. It is much more convenient not to bother to take the square root, so let us talk about the square of the distance; that is, let us find out whether

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2. \tag{1.17}$$

It had better be—and if we substitute Eq. (1.5) we do indeed find that it is. So we see that there are other kinds of equations which are true for any two coordinate systems.